# Haralick Texture Features

## Notation

$N$: The number of gray levels in the image (dimensions of the GLCM).

$p(i,j)$: The normalized probability of co-occurrence for gray levels $i$ and $j$:

$$p(i,j) = \frac{P(i,j)}{\sum_{i=1}^{N}\sum_{j=1}^{N} P(i,j)}$$

$\mu_x, \mu_y$: The marginal means of the rows and columns:

$$\mu_x = \sum_{i=1}^{N} i \sum_{j=1}^{N} p(i,j) \quad ; \quad \mu_y = \sum_{j=1}^{N} j \sum_{i=1}^{N} p(i,j)$$

$\sigma_x, \sigma_y$: The marginal standard deviations:

$$\sigma_x = \sqrt{\sum_{i=1}^{N}(i-\mu_x)^2 \sum_{j=1}^{N} p(i,j)} \quad ; \quad \sigma_y = \sqrt{\sum_{j=1}^{N}(j-\mu_y)^2 \sum_{i=1}^{N} p(i,j)}$$

$k$: An index variable used for sum $(2 \ldots 2N)$ and difference $(0 \ldots N-1)$ distributions.

$p_{x+y}(k)$: The probability distribution of the sum of gray levels:

$$p_{x+y}(k) = \sum_{i=1}^{N}\sum_{j=1}^{N} p(i,j) \quad \text{where } i+j=k$$

$p_{x-y}(k)$: The probability distribution of the difference of gray levels:

$$p_{x-y}(k) = \sum_{i=1}^{N}\sum_{j=1}^{N} p(i,j) \quad \text{where } |i-j|=k$$

$\mu_{x+y}$: The mean of the sum distribution $p_{x+y}(k)$:

$$\mu_{x+y} = \sum_{k=2}^{2N} k \cdot p_{x+y}(k)$$

$\mu_{x-y}$: The mean of the difference distribution $p_{x-y}(k)$:

$$\mu_{x-y} = \sum_{k=0}^{N-1} k \cdot p_{x-y}(k)$$

$HX, HY$: The entropies of the marginal distributions $p_x$ and $p_y$:

$$HX = -\sum_{i=1}^{N} p_x(i)\log(p_x(i)) \quad ; \quad HY = -\sum_{j=1}^{N} p_y(j)\log(p_y(j))$$

$HXY$: The joint entropy:

$$HXY = -\sum_{i=1}^{N}\sum_{j=1}^{N} p(i,j)\log(p(i,j))$$

$HXY1$:

$$HXY1 = -\sum_{i=1}^{N}\sum_{j=1}^{N} p(i,j)\log(p_x(i)p_y(j))$$

$HXY2$:

$$HXY2 = -\sum_{i=1}^{N}\sum_{j=1}^{N} p_x(i)p_y(j)\log(p_x(i)p_y(j))$$

# 1 Autocorrelation

[1], [2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}(i\cdot j)p(i,j)$$

Put simply, autocorrelation gives the sum of the products of expected vlaues $\big(E[p(i,j)] = E[p_x(i)]E[p_y(j)] = (i\cdot j)p(i,j)\big)$. If we assume $N$ to be held constant, the distribution of the GLCM entries heavily determine the value of this feature. So if an image is relatively brighter autocorrelation will return a higher value.

As i or j increase the function increases. In terms of GLCMs as the indicies distance from the (1,1) entry increases the higher the wieghting of that entry; This means dark pixels co-occur with dark pixels, or low-intensity regions are dominant. The most bottom right entry $(N, N)$, has the highest possible weight. This means bright pixels co-occur with bright pixels. Also The top right and bottom left corners both have higher weights. In the image bright pixels co-occur with bright pixels.

# 2 Cluster Prominence

[2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}(i+j-\mu_x-\mu_y)^4 p(i,j)$$

in [1] There is a typo, cluster prominence is raised to the third power, instead it should be raised to the fourth as per other sources. Also, $-\mu_x - \mu_y$ is written as $-2\mu$ where $\mu = (\mu_x + \mu_y)/2$.

$\mu_x$ and $\mu_y$ are the mean gray levels of the rows and columns of the GLCM. They represent the expected intensity co-occurrence in the matrix.

The wieght can be rewritten as $(i-\mu_x)+(j-\mu_y)$, the sum of deviations from the mean. This term measures the distance of a specific co-occurrence $(i, j)$ from the mean. The sum $(i + j)$ is compared against the sum of the means $(\mu_x + \mu_y)$, so this feature focuses on how much the texture "protrudes" from the average. Since the difference is raised to the 4th power, it heavily penalizes entries that are far away from the center of the GLCM. This would also ensure a positive value $\big((-a)^4 = a^4\big)$

Cluster prominence is a measure of the asymmetry of the GLCM. A High cluster prominence indicates the image has disturbances, where patterns occur that is different from usual co-occurrences. A low cluster prominence indicates that the GLCM is concentrated near the mean (the mean can be anywhere).

# 3 Cluster Shade

[2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}(i+j-\mu_x-\mu_y)^3 p(i,j)$$

Typo in [1], mismatched with cluster prominence.

Similar to the interpetation of cluster prominence; except that this function is raised to the third power. This allows for negative values. If a co-occurnce with gray levels darker than the mean $(i+j<\mu_x+\mu_y)$, this produces a negative weight, thus reduce the sum. So if the mean is relatively dark (most $i+j$ will be greater than the mean, if these gray levels exist in the image), the rest of the possible co-occurrences will be brighter, increasing the sum. This will likely be higher in GLCMs with concentrations near the $(1,1)$ entry, i.e. darker images. As opposed to Cluster prominence and cluster tendency, this feature dosen't drop when the GLCM is concentrated around the mean.

# 4 Cluster Tendency

[2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}(i+j-\mu_x-\mu_y)^2 p(i,j)$$

Similar to the interpetation of cluster prominence.

## 4.1 Sum Variance

[1], [2], [3] (Mathematically equivalent to cluster tendency)

$$\sum_{k=2}^{2N}(k-\mu_{x+y})^2 p_{x+y}(k)$$

# 5 Contrast

[2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}(i-j)^2 p(i,j)$$

The term $(i-j)^2$ acts as a weighting kernel that measures the squared distance from the main diagonal of the GLCM. When $i-j=0$, the weighting is zero. These entries represent pixels co-occurrences with identical gray level. The weighting increases quadratically as the distance $|i-j|$ increases. Consequently, this feature is highly sensitive to entries in the far corners of the matrix which represent transitions between extreme dark and extreme bright pixels.

If the probability mass of $p(i,j)$ is concentrated on or near the main diagonal, the summation remains small. In an image this translates to a homogeneous image where neighboring pixels have very similar gray levels, less intense edges/transitions.

If the mass is widely distributed across the off-diagonals $(|i-j|\to N)$, the contrast value grows. This indicates more frequent large intensity transitions. Which, in an image, can corresponds to sharp edges, high-frequency noise, or coarseness.

# 6 Correlation

[2], [3]

$$\frac{\sum_{i=1}^{N}\sum_{j=1}^{N}(i\cdot j)p(i,j)-\mu_x\mu_y}{\sigma_x(i)\sigma_y(j)}$$

in [1], its written as:

$$\sum_{i=1}^{N}\sum_{j=1}^{N}(\frac{i-\mu_x}{\sigma_x})(\frac{j-\mu_y}{\sigma_y})p(i,j)$$

This is similar to a discrete implementation of the Pearson correlation coefficient for the joint probability mass function $p(i,j)$. It measures how much the gray level of a pixel is linearly predictable from its neighbor.

Entries where both $i$ and $j$ are simultaneously greater than or less than the mean are positive thus . This feature favors . Typically if the contrast of an image is high, the correlation would be low (not necessarily vice versa). And similar to cluster prominence, they pick up on deviance from the mean of the GLCM (but correlation is minimized by high intensity adjacencies instead of concentration around the mean).

Since the terms are standardized by $\sigma_x$ and $\sigma_y$, this feature is bounded (with respect to the GLCM).

# 7 Difference Average

[2]

$$\sum_{k=0}^{N-1} k \cdot p_{x-y}(k)$$

Similar to the interpretation of Contrast.

## 7.1 Dissimilarity

[1], [2], [3] (mathematically equivalent to Difference Average)

$$\sum_{i=1}^{N}\sum_{j=1}^{N}|i-j|p(i,j)$$

# 8 Difference Entropy

[2]

$$\sum_{k=0}^{N-1} p_{x-y}(k)\log_2\left(p_{x-y}(k)+\epsilon\right)$$

in [3],

$$\sum_{i=0}^{N-1} p_{x-y}(i)\log_2\left(p_{x-y}(i)\right)$$

in [1],

$$\sum_{k=0}^{N-1} p_{x-y}(k)\log\left(p_{x-y}(k)\right)$$

Shannon Entropy applied to the distribution of gray-level differences ($p_{x-y}(k)$). Simply Difference Entropy quantifies the complexity and disorder of the local intensity changes rather than the intensity of the changes themselves. This feature uses a marginalization of the GLCM along its off-diagonals. It measures the uniformity of the distribution of differences. If the probability mass is concentrated on just a few values of $k$, the entropy is low. If the mass is distributed across all possible differences, entropy is maximized.

At a high difference entropy value the image contains a vast variety of local transitions (some subtle, some sharp) with no dominant pattern. There is high uncertainty in predicting the difference between a pixel and its neighbor. At a low difference entropy value the image has order or structural simplicity. The distribution of differences is sparse.

## 9 Difference Variance

[1], [2]

$$\sum_{k=0}^{N-1} (k - \mu_{x-y})^2 p_{x-y}(k)$$

This feature measures the variance of the marginalized difference distribution $p_{x-y}(k)$. The term $(k - \mu_{x-y})^2$ acts as a weighting function that penalizes deviations from the mean difference $\mu_{x-y}$. $\mu_{x-y}$ represents the mean intensity classes (each $i - j$ is a class). The off-diagonal farthest from the main diagonal represents the class of highest intensity. If an image has a very consistent texture where the intensity change between neighbors is stable, $p_{x-y}(k)$ will be highly concentrated around its mean, resulting in a low variance.

Low difference variance suggests a smooth texture. Even if the image is high-contrast, if the transitions between light and dark pixels follow a regular, repetitive rule, the variance of those differences remains small.

## 10 Energy

[2], [3] Also knwon as 'Joint Energy' and 'Angular Second Moment'.

$$\sum_{i=1}^{N} \sum_{j=1}^{N} p(i, j)^2$$

Energy is a measure of the concentration of the probability distribution within the GLCM. This feature operates on the principle that the sum of squares of a set of probabilities is maximized when the probability mass is concentrated in a few entries and minimized when it is spread uniformly. If an image is dominated by a certain co-occurnce, wether $i = j$ (a singulare gray level) or $i \neq j$ (a singular gradient), the normalized GLCM at entry $(i, j)$ will in turn have a large probability. This concentration (energy) at a singular entry causes the feature to increase; concentrations at multiple entries can produce a high feature value as long as it's probabilities are much larger than the rest.

## 11 Entropy

[2] Also known as 'Joint Entropy'.

$$-\sum_{i=1}^{N} \sum_{j=1}^{N} p(i, j) \log_2 \big(p(i, j) + \epsilon\big)$$

[3]

$$-\sum_{i=1}^{N} \sum_{j=1}^{N} p(i, j) \log_2 \big(p(i, j)\big)$$

[1]

$$-\sum_{i=1}^{N} \sum_{j=1}^{N} p(i, j) \log \big(p(i, j)\big)$$

Shannon entropy applied to the normalized glcm. Entropy is maximized when the distribution is uniform ($p(i, j) = 1/N^2$ for all $i, j$), meaning every possible gray-level transition occurs with equal frequency. Entropy is minimized when the image is perfectly homogeneous, and only one entry in the GLCM has a probability of 1.

In the context of a GLCM, entropy quantifies the structural unpredictability of gray level co-occurrences. A low entropy GLCM indicates that only a few gray level pairs occur with significant probability (typical of highly regular textures, large uniform regions, or consistent gradients). A high entropy GLCM reflects a broad spread of co-occurrence probabilities, corresponding to complex, irregular, or noisy textures.

Entropy is agnostic to where the mass lies in the matrix (diagonal vs. off-diagonal). This feature can play as a compliment of Energy and Homeginiety-like features feature.

# 12 Homogeneity 1

[1], [2], [3] Also known as 'Inverse Difference'

$$\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{p(i,j)}{1+|i-j|}$$

Compared to Inverse difference moment, the denominator grows linearly rather than quadratically, Inverse difference doesn't nullify off-diagonal probabilities as quickly. This results in a feature that maintains a higher value even in the presence of low-level noise.

## 12.1 Inverse Difference Normalized

[2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{p(i,j)}{1+\frac{|i-j|}{N}}$$

# 13 Homogeneity 2

[2], [3] Also known as 'Inverse Difference Moment'

$$\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{p(i,j)}{1+|i-j|^2}$$

In [1], it is titled Homogeneity.

This feature quantifies the degree of local similarity in an image. The denominator term $1+(i-j)^2$, acts as a penalty that is minimized when $i=j$ (the diagonal entries). In a region where pixel co-occurences are nearly identical, $(i-j)\to 0$, the weights approach 1. In a GLCM the diagonals and near diagonals will have the most contribution to the sum. As the gradients $(i-j)$ increase, the denominator grows quadratically which reduces the contribution of those entries to the total sum. This feature can be seen as an inverse or compliment of contrast.

## 13.1 Inverse Difference Moment Normalized

[2], [3]

$$\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{p(i,j)}{1+\frac{|i-j|^2}{N^2}}$$

# 14 Information measure of correlation 1

[2], [3]

$$\frac{HXY-HXY1}{\max\{HX,HY\}}$$

# 15 Information measure of correlation 2

[2], [3]

$$\sqrt{1-e^{-2(HXY2-HXY)}}$$

## 16 Inverse Variance

[2]

$$\sum_{k=1}^{N-1} \frac{p_{x-y}(k)}{k^2}$$

[3]

$$\sum_{i=1}^{N} \sum_{j=1}^{N} \frac{p(i,j)}{|i-j|^2} \ , \quad i \neq j$$

The $\frac{1}{|i-j|^2}$ is similar to Inverse difference moment. Instead this feature begins at $k = 1$ to avoid divition by 0. Because $k^2$ grows quadratically, the weights $1/k^2$ decreases quickly as we move closer to the diagonal. For example, a difference of $k = 1$ has a weight of 1.0, while a difference of $k = 10$ has a weight of 0.01. This feature has stronger wieghts for off-diagonals further from the diagonal, but instead of factoring by $k$ (as done in Contrast) it is factored by $1/k^2$.

High Inverse Variance suggests a homogeneous image where majority of the neighboring pixels have identical or nearly identical gray levels. (plots of Inverse variance and contrast may seam identical because only two gray levels are used in our experiments).

## 17 Joint Average

[2]

$$\sum_{i=1}^{N} \sum_{j=1}^{N} i \cdot p(i,j)$$

Use only on symmetric GLCMs.

This feature focuses on each gray levelJoint Average acts as a proxy for the local brightness level of the textured region. This feature is half of the Sum Average. Similar to Sum average, but here the whole row of a GLCM has the same weight.

## 18 Maximal Correlation Coefficient

[1], [2]

$$\sqrt{\lambda_2 Q(i,j)}$$

$$Q(i,j) = \sum_{k=0}^{N} \frac{p(i,k)p(j,k)}{p_x(i)p_y(k)}$$

## 19 Maximum Probability

[2], [3]

$$\max\big(p(i,j)\big)$$

This feature simply returns the largest probability of co-occurences from a GLCM. This feature gives less information about the texture, instead it may indicate if an image is dominated by a certain co-occurence.

## 20 Sum Average

[2]

$$\sum_{k=2}^{2N} k \cdot p_{x+y}(k)$$

[3]

$$\sum_{i=2}^{2N} i \cdot p_{x+y}(i)$$

This gives us the sum of the indices weighted by the k value. Sum Average is essentially calculating the center of mass along the axis perpendicular to the main diagonal (the $i+j$ lines). It considers pixels that actually participate in the co-occurrence relationship (in a cross diagonal there is a certain set of co-occurences). Higher weight is given to brighter values, like autocorrelation, except that this feature lacks weight in the cross corners.

e.g The cross-diagonal that intersects the $(3,3)$ entry of the GLCM, $4 \leq k \leq 6$ The faeture compares the average within each cross diagonal (instad of the average of the whole GLCM), this captures variances with all the co-occurences where the gray level co-occurs with gray level $= 3$. But gray levels greater than 3 arent captured.

# 21 Sum Entropy

[2]

$$\sum_{k=2}^{2N} p_{x+y}(k) \log_2 \left(p_{x+y}(k) + \epsilon\right)$$

[3]

$$-\sum_{i=2}^{2N} p_{x+y}(i) \log_2 \left(p_{x+y}(i)\right)$$

[1]

$$-\sum_{k=2}^{2N} p_{x+y}(k) \log \left(p_{x+y}(k)\right)$$

Sum Entropy is then the Shannon Entropy of the $p_{x+y}$ distribution. Similar to Entropy, if the distribution is concentrated the Sum Entropy value will be near zero, adn grows as the distribution is more uniform.

This feature can be used to describe the spatial distribution of brightness in a texture. High value indicates a texture where many different combinations of total brightness exist across the image. This is typical of heterogeneous or "noisy" environments where light and dark areas are mixed in many different ways.

# 22 Sum of Squares

[2]

$$\sum_{i=1}^{N} \sum_{j=1}^{N} (i - \mu_x)^2 p(i,j)$$

[1], [3] Denoted as 'Variance':

$$\sum_{i=1}^{N} \sum_{j=1}^{N} (i - \mu)^2 p(i,j)$$

# References

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